

Predicting Interaction

Toward a Predictive Theory of Computational Interaction

Formal research paper

Abstract

Computational interaction is predictively tractable only under declared targets, horizons, observation models, populations, interventions, and loss functions. Existing research supplies successful local predictive laws and models: Fitts' law predicts constrained movement time; Hick-Hyman relations predict choice latency under controlled alternatives; GOMS and the Model Human Processor predict skilled task time; cognitive architectures generate task-specific performance trajectories; Bayesian, survival, sequential, and machine-learning models forecast events under uncertainty. These successes do not establish a universal predictive theory. Discovery, errors, recovery, abandonment, adaptation, expertise, transfer, and strategy differ in observability, temporal scale, causal dependence, and distributional stability. Prediction must therefore be typed.

This paper distinguishes forecasting from explanation, retrodiction, diagnosis, causal effect estimation, and intervention planning. It proposes a prediction ontology and a layered framework in which predictions are conditional distributions over declared interaction targets generated from episode state, opportunity structure, projection, behavioral history, agent experience, strategy hypotheses, and context. Predictive claims must report calibration, discrimination, decision utility, uncertainty, transport conditions, and failure criteria. The central conclusion is that computational interaction supports predictive science, but only as a family of scoped, probabilistic, and updateable models. Low-level performance and near-term transitions are often strongly predictable; higher-order constructs and long-horizon outcomes are forecastable only as distributions and frequently remain unidentified without intervention or richer observation.

1. Problem Statement

The preceding research program defines interaction episodes, behavioral traces, opportunity profiles, strategies, transfer, similarity, and a measurement architecture. Prediction is the next test of whether these constructs support empirical science rather than only conceptual organization. The relevant question is not whether some interaction outcomes correlate with earlier variables. It is whether future or unobserved interaction variables can be forecast with known uncertainty under declared conditions and whether the resulting forecasts remain valid across participants, tasks, projections, time, and interventions.

Interaction is an especially difficult predictive domain. The system and agent mutually constrain later trajectories; observations are partial; behavior changes beliefs and opportunity; participants learn; interfaces and policies change; and the prediction itself may alter behavior. Prediction targets also occupy different levels. A movement time is directly measurable, while a strategy, expertise state, or believed action set is latent. A model can predict a trace without explaining its mechanism, explain a mechanism without predicting held-out traces, or estimate an intervention effect without accurately predicting individual outcomes.

The central research problem is therefore to define a disciplined predictive architecture that separates target types, horizons, information sets, causal questions, uncertainty, and validity domains. The objective is not one universal model. It is the minimum framework required to state, test, compare, update, and falsify predictive claims about computational interaction.

2. Literature Review Method

The review was organized by predictive function rather than by discipline. Primary and foundational sources were prioritized for: statistical distinctions between explanation and prediction; human-performance laws; engineering models of skilled interaction; cognitive architectures; Bayesian and sequential models; machine learning; survival and event-history analysis; causal inference; transfer and transport; and predictive uncertainty. The earlier Interaction, Behavior, Measurement, Opportunity, Similarity, Transfer, Strategy, and Formal Models papers were treated as constraints on the predictive target and evidence model, not as evidence that prediction must succeed.

Each tradition was evaluated against five questions: What is predicted? From what information? At what horizon and unit? Under what assumptions? By what evidence is success or failure established? Claims were rejected as

predictive when they were evaluated only in-sample, used future information, relied on post-outcome labels, lacked a baseline, or did not report uncertainty and calibration.

3. Explanation, Prediction, and Intervention

Prediction and explanation are distinct scientific aims. Predictive modeling minimizes expected loss for unobserved cases under a specified data-generating regime. Explanatory modeling aims to identify or support mechanisms, causes, or theoretical dependencies. A structurally faithful explanatory model can predict poorly when unmodeled variability dominates, while a flexible predictor can forecast accurately without identifying the generating mechanism. This distinction is emphasized by Shmueli and by Yarkoni and Westfall.

Four additional functions must remain separate:

- Retrodiction estimates an unobserved or hidden past variable from later evidence.
- Diagnosis classifies or explains a current condition from observations.
- Causal effect estimation predicts the distribution that would follow an intervention rather than passive observation.
- Decision or control selects an intervention using predicted consequences and utilities.

Observational prediction estimates $P(Y_{\text{future}} | H_{\text{observed}})$. Intervention prediction estimates $P(Y_{\text{future}} | \text{do}(A=a), H_{\text{observed}})$. The latter cannot generally be obtained from the former without causal assumptions, identification conditions, or experimental variation. A system that accurately predicts abandonment from frustration signals does not thereby establish that reducing those signals will prevent abandonment.

4. Prediction Ontology

Concept	Definition	Example
Prediction target	The future, hidden, or counterfactual variable to be forecast.	Discovery event, error, completion time, strategy class.
Prediction unit	Entity over which the prediction is defined.	Action, episode, person-task pair, team, system deployment.
Horizon	Temporal or structural distance between information cutoff and target.	Next action; remainder of episode; next session.
Information set	Variables legally available at prediction time.	State, projection, opportunity, prior trace, experience.
Predictor	Function or probabilistic model mapping information to a predictive distribution.	Hazard model, POMDP belief model, neural sequence model.
Predictive distribution	Uncertainty-bearing estimate over outcomes.	Probability of error; interval for completion time.
Decision threshold	Rule converting a distribution into an action or label.	Intervene when failure probability exceeds 0.7.
Loss or utility	Cost function defining predictive quality for a use.	Log loss, time error, false-negative cost.
Validity domain	Population, task, projection, context, and time for which evidence supports use.	Trained novices on desktop form completion.
Transport condition	Assumption under which performance generalizes to a target domain.	Stable measurement, mechanism, or conditional distribution.
Calibration	Agreement between predicted probabilities and observed frequencies.	Among 0.8 predictions, about 80% occur.
Discrimination	Ability to rank or separate outcomes.	AUC, concordance, precision-recall.
Sharpness	Concentration of a calibrated predictive distribution.	Narrow intervals with valid coverage.
Prediction failure	Violation of accuracy, calibration, coverage, utility, or scope.	Drift, subgroup failure, intervention mismatch.

5. What Is Predictable?

Predictability is not a property of a phenomenon alone. It is a relation among target, horizon, information set, population, observation process, model class, and tolerance. The same outcome may be highly predictable at the population level and weakly predictable for an individual. Aggregate movement time may follow a stable law while a particular person's next action remains uncertain.

Target	Expected predictability	Suitable models	Principal limits
Low-level motor/perceptual performance	Often high under controlled geometry, practiced action, and stable devices.	Fitts-type movement time; Hick-Hyman choice latency.	Device variability, strategy changes, fatigue, accessibility, nonstationarity.
Skilled procedural task time	Moderate to high for routine, error-free, well-specified tasks.	KLM/GOMS and Model Human Processor.	Learning, interruptions, errors, interpretation, collaborative work.
Next action or transition	Moderate when state, opportunity, policy, and recent history are observed.	Sequence models, Markov/POMDP models.	Hidden goals, multiple valid strategies, changing opportunity.
Discovery	Probabilistic and context-sensitive.	Hazard of first discovery from exposure, signaling, search history.	Unobserved prior knowledge and attention.
Errors	Useful risk prediction possible; exact event prediction limited.	Human reliability, classification, state-space risk models.	Rare events, adaptive avoidance, reporting bias.
Recovery	Predictable conditional on error type, visible recovery paths, belief, and skill.	Competing-risk or sequential models.	Unobserved interpretation and external consequences.
Abandonment	Often forecastable as a time-to-event risk.	Survival analysis using friction and progress signals.	Censoring, return behavior, changing goals.
Adaptation	Predictable as a distribution over learning curves or policy changes.	Bayesian learning, power-law/exponential learning models.	Task change, exploration, motivation, strategy shifts.
Expertise	Population-level classification is possible; individual developmental trajectory is weakly identified.	Skill models, cognitive architectures, longitudinal models.	Domain specificity and sparse longitudinal data.
Transfer	Conditional and asymmetrical; requires source-target relation.	Hierarchical or transport models.	Negative transfer, representation mismatch, context dependence.
Strategy	Latent; only probabilistic inference from multiple traces and interventions is defensible.	Inverse planning, mixture models, inverse reinforcement learning.	Observational equivalence and strategy switching.
Long-horizon interaction trajectory	Generally low without closed-world assumptions and continual updating.	Scenario distributions, policy-conditioned simulation.	Compounding uncertainty, exogenous change, reflexivity.

6. Predictive Traditions

6.1 Human-performance laws

Fitts' law and Hick-Hyman relations demonstrate that constrained components of interaction can support compact, quantitative prediction. Their scientific value comes from declared task structure, measurable variables, reproducible relations, and explicit residual variation. Their limitation is scope: they predict selected performance dimensions under particular action and choice structures, not whole interactions.

6.2 Engineering models of interaction

GOMS, the Keystroke-Level Model, and the Model Human Processor predict skilled, largely error-free task performance from decomposed operators and processing assumptions. They are strongest when methods and operators are known and weakest when discovery, learning, interpretation, collaboration, interruption, or strategy change dominates.

6.3 Cognitive architectures

ACT-R, EPIC, and related architectures generate predictions by instantiating mechanistic assumptions about memory, perception, cognition, and action. They can explain and predict within modeled tasks, but performance depends on task encoding, model construction, parameter selection, and whether the architecture captures the relevant environmental coupling. Architecture-level commitments do not remove the need for out-of-sample validation.

6.4 Bayesian and sequential models

Bayesian models express uncertainty over states, parameters, hypotheses, and future observations. State-space models, hidden Markov models, MDPs, and POMDPs are natural for interaction because they distinguish latent state, observation, action, transition, and belief. They become predictive only after the transition and observation models are estimated or constrained, and their validity depends on stationarity and state sufficiency.

6.5 Machine learning

Machine learning expands predictive capacity through flexible representation and regularized out-of-sample optimization. It is appropriate for heterogeneous traces, nonlinear interactions, and high-dimensional histories. It

does not automatically provide causal interpretation, transportability, or calibrated uncertainty. Strong performance within a random split may conceal failure across users, tasks, time, systems, or interventions.

6.6 Survival, reliability, and event-history models

Discovery, failure, recovery, and abandonment are naturally represented as time-to-event or competing-risk problems. These models handle censoring and changing hazards more appropriately than binary labels. Recurrent-event and multi-state variants can represent repeated errors, recovery cycles, and episode transitions.

7. Candidate Predictive Framework

Let an interaction episode be represented by the layered formal model established previously. For participant x at time t , define the legally available predictive information:

$$H_t = \langle s_t, \pi_t, O_t, Q_t, y_{0:t}, e_x, b_t, z_t, c_t \rangle$$

where s_t is system state; π_t is projection; O_t is typed opportunity structure; Q_t is authority and policy state; $y_{0:t}$ is the recorded and attributed trace up to the cutoff; e_x is measured experience; b_t is a belief-state estimate where justified; z_t is a strategy hypothesis distribution; and c_t is declared context.

For target Y at horizon h under prediction regime r , the predictor produces:

$$p(Y_{t+h} | H_t, r, M, D)$$

where M identifies model assumptions and D identifies the training or calibration evidence. Under an intervention a , the target becomes $p(Y_{t+h} | \text{do}(a), H_t, M, D)$. The two expressions must not be substituted without identification assumptions.

The predictive framework contains six linked layers:

- Target layer: target, unit, horizon, outcome definition, competing outcomes, and censoring.
- Information layer: cutoff, observable variables, derived variables, latent estimates, and prohibited future information.
- Model layer: predictor family, priors or regularization, update rule, and causal assumptions.
- Uncertainty layer: aleatoric, epistemic, measurement, distributional, and structural uncertainty.
- Evaluation layer: baselines, held-out design, calibration, discrimination, interval coverage, and utility.
- Validity layer: population, task, projection, time, intervention, and monitoring conditions.

8. Causal Assumptions

Predictive accuracy does not establish a causal mechanism. Causal claims require an explicit causal graph or structural model, temporal ordering, intervention semantics, and assumptions concerning confounding, selection, measurement, and interference. Four distinctions are mandatory:

- Prediction under observation versus prediction under intervention.
- Individual treatment or intervention response versus average effect.
- Mediation as a causal claim versus predictive feature contribution.
- Transport across contexts versus ordinary cross-validation within one context.

The interaction domain intensifies interference: one participant's action changes the state and opportunities of others; system adaptations alter later observations; and predictive interventions may change the target process. Static independent-and-identically-distributed assumptions are therefore often inappropriate. Sequential causal models, randomized interventions, natural experiments, or explicit policy evaluation are required for intervention claims.

Claim type	Target	Required assumptions	Example
Associational forecast	Future outcome under the observed regime.	Temporal ordering; stable conditional distribution.	Will the current episode end in abandonment?
Causal effect	Outcome distribution under an	Identification, consistency,	Would adding a recovery signifier

	intervention.	exchangeability or randomized design.	reduce failure?
Policy value	Expected utility of an adaptive intervention policy.	Sequential ignorability or experimental policy data; positivity.	When should help be offered?
Mechanism prediction	Intermediate states generated by a theory.	Correct structural specification and measurable mediators.	Will belief update precede selection?
Transported prediction	Outcome in a new population or projection.	Declared invariant mechanisms or reweighting assumptions.	Will a model trained on GUI sessions work for voice?

9. Uncertainty and Prediction Limits

A prediction should be a distribution, interval, set, ranked list, or hazard unless deterministic structure is independently established. Point predictions conceal uncertainty and invite false precision. The framework distinguishes:

- Aleatoric uncertainty: irreducible variation conditional on the model and information set.
- Epistemic uncertainty: uncertainty about parameters, functions, or hypotheses due to limited evidence.
- Measurement uncertainty: error introduced by instruments, segmentation, coding, and construct operationalization.
- State uncertainty: partial observability of current system, agent, or social state.
- Structural uncertainty: uncertainty over causal structure, model family, or latent-state representation.
- Distributional uncertainty: uncertainty caused by population, task, projection, policy, or temporal shift.
- Reflexive uncertainty: change caused by revealing or acting on a prediction.

Conformal prediction can provide finite-sample marginal coverage under exchangeability, but it does not solve conditional validity, causal transport, or nonstationary sequential interaction. Bayesian credible intervals depend on model and prior assumptions. Ensembles estimate some forms of model uncertainty but do not guarantee coverage. Uncertainty claims must therefore name the guarantee and its assumptions.

Fundamental limits arise from observational equivalence, hidden goals, underdetermined strategies, partial observability, chaotic or highly sensitive dynamics, changing policies, endogenous learning, rare outcomes, insufficient support for interventions, and open-world exogenous events. In such cases the scientifically correct prediction may be a broad set, a ranked scenario distribution, or a declaration of non-identifiability.

10. Target-Specific Analysis

10.1 Discovery

Discovery should be predicted as a time-to-first-event process relative to an opportunity contract. Predictors may include exposure, signaling, search path, prior experience, modality, task state, and prior failed attempts. Unselected available actions cannot be labeled undiscovered without an independent awareness measure. Competing events include successful discovery, prompted discovery, alternate-path completion, and abandonment.

10.2 Opportunity

System-relative feasible, permitted, enabled, and exposed sets are often computable rather than statistically predicted when the model is complete. Agent-relative believed actions and expected consequences require probabilistic estimation. Predictions should distinguish changes in actual opportunity from changes in belief or signaling.

10.3 Errors

Error prediction should define the normative or task criterion, error opportunity, and exposure time. Rare-event evaluation requires precision-recall, calibrated risk, and cost-sensitive metrics rather than accuracy alone. Error labels must not collapse execution failure, goal mismatch, policy denial, misunderstanding, and system fault.

10.4 Recovery

Recovery prediction is conditional on an initiating disruption and the available recovery graph. Suitable targets include time to repair, selected recovery path, probability of escalation, and residual consequence. Believed reversibility and actual reversibility must remain separate.

10.5 Abandonment

Abandonment is a censored event, not merely absence of completion. Models should distinguish temporary pause, session termination, task substitution, and permanent non-return. Survival models with time-varying covariates are preferable when the hazard changes with progress and friction.

10.6 Adaptation

Adaptation prediction concerns change in policy, belief, skill, or performance as experience accumulates. Learning curves forecast aggregate change but can conceal strategy shifts. Hierarchical state-space models can represent individual variation and update online.

10.7 Expertise

Expertise is multidimensional and domain-relative. Prediction may classify current skill or forecast performance under familiar tasks, but long-term expertise development requires longitudinal exposure, practice quality, feedback, motivation, and domain structure. Demographic variables should not stand in for measured experience.

10.8 Transfer

Transfer prediction requires a source-target similarity contract and a carrier hypothesis. Predictions should estimate positive, null, or negative source effects and uncertainty. Ordinary similarity or source performance alone is insufficient.

10.9 Strategy

Strategy is latent and typically non-identifiable from one trace. Predictive inference should return a posterior over strategy hypotheses or next-action distributions, test alternative models, and update when contingencies reveal discriminating behavior. A model that predicts actions well may still misidentify strategy.

11. Prediction Protocol

Step	Requirement
1. Declare target	Define the outcome, unit, horizon, event time, censoring, and competing outcomes.
2. Freeze information cutoff	List every variable available at prediction time; prohibit future-derived features.
3. State use	Separate scientific forecasting, diagnosis, causal intervention, control, and operational triage.
4. Specify population and context	Declare users or agents, task, projection, device, authority, environment, and time period.
5. Register baselines	Include prevalence, persistence, simple rules, human judgment, and established task models.
6. Select evaluation design	Use temporal, participant, task, projection, or deployment holdout appropriate to the intended claim.
7. Fit without leakage	Keep preprocessing, feature selection, imputation, and tuning within training folds.
8. Quantify uncertainty	Produce calibrated probabilities, intervals, sets, or hazards with named guarantees.
9. Evaluate multiple properties	Report discrimination, calibration, coverage, sharpness, utility, subgroup and context performance.
10. Test transport	Evaluate across time, participants, tasks, projections, systems, and policies when those claims are intended.
11. Test interventions separately	Use experiments or identified causal models for claims about changing outcomes.
12. Monitor and update	Detect drift, recalibrate, revise the validity domain, and document model changes.

13. State failure conditions	Define minimum acceptable gain, calibration, coverage, and decision utility before use.
14. Preserve evidence lineage	Link each prediction to the interaction, behavior, and measurement contracts that generated its inputs.

12. Evaluation and Falsification

A predictive theory of interaction should be rejected or narrowed if its constructs provide no reproducible out-of-sample gain over simpler baselines, if uncertainty is systematically miscalibrated, or if performance collapses under the contexts included in its claimed validity domain.

Claim	Test	Falsifying or narrowing evidence
P1 Scoped predictability	Declared interaction targets can be predicted above preregistered baselines.	Repeated external tests show no meaningful gain.
P2 Typed advantage	Opportunity, trace, transfer, and strategy variables improve target-specific predictions when relevant.	Simpler state and history features match or outperform them.
P3 Uncertainty validity	Prediction intervals or probabilities are calibrated within declared scope.	Persistent undercoverage or probability miscalibration.
P4 Transport	Specified invariant structures support prediction across declared contexts.	Performance degrades beyond tolerance under planned shifts.
P5 Intervention separation	Observational and interventional predictions differ where confounding or feedback exists.	Observed association reliably equals intervention effect under tested conditions, narrowing the distinction.
P6 Prediction-explanation distinction	Predictive ranking and mechanistic adequacy can diverge.	Across broad tests, the best explanatory model always dominates prediction, contrary to current evidence.
P7 Strategy uncertainty	Multiple strategies remain plausible under observationally equivalent traces.	Discriminating evidence reliably identifies one strategy from the available trace alone.
P8 Updateability	Sequential updating improves calibration under learning and drift.	Static models remain equally valid across declared nonstationarity.

Evaluation must include out-of-sample prediction. In-sample fit, significance, explained variance, and post hoc reconstruction do not suffice. For probabilistic classification, report proper scoring rules and calibration in addition to threshold metrics. For time-to-event prediction, report time-dependent discrimination, calibration, and censoring-aware scores. For sequences, report next-event and multi-step performance with horizon-specific degradation. For decisions, report expected utility and harm under the operational threshold.

13. Threats to Validity

- Target leakage can enter through post-outcome fields, retrospective labels, or segmentation choices informed by the full trace.
- Random train-test splits can overstate generalization when the intended use concerns new people, tasks, projections, or future periods.
- Construct labels such as confusion, expertise, and strategy may be unreliable or circular.
- Adaptive systems violate stationarity because predictions and interventions change later data.
- Rare failures can produce misleading accuracy and unstable subgroup estimates.
- Models may encode historical access, training, or policy inequalities and treat them as stable individual propensities.
- Predictive success can invite unwarranted causal and psychological interpretation.
- Closed-world simulations may omit exogenous events that dominate deployed interaction.
- Aggregate calibration can coexist with severe conditional or subgroup miscalibration.
- A benchmark can become part of the design process, causing overfitting to the evaluation regime.

14. Implications for Computational Interaction Science

Prediction provides a stronger test of the research program than conceptual coherence alone. The established constructs should earn their place by improving forecast quality, uncertainty, transport, or intervention design relative to simpler state-and-trace models. Typed opportunity variables should help predict discovery and invalid attempts; transfer variables should improve target-context forecasts; strategy distributions should improve

contingent action prediction; and formal interaction models should constrain impossible transitions and support simulation.

Predictive success would not by itself establish an independent discipline. Many fields develop predictive models of interaction. A discipline-level claim would require cumulative laws or mechanisms, stable validity domains, shared benchmarks, transportable models, and explanatory or intervention gains that are not adequately supplied by existing HCI, human factors, psychology, machine learning, control, or systems theory.

The immediate empirical program should therefore prioritize preregistered, multi-context forecasting studies with frozen contracts and strong baselines. The decisive evidence is incremental, held-out performance and calibration—not the apparent completeness of the ontology.

15. Conclusion

Computational interaction supports predictive scientific theories, but only in a scoped and plural sense. There is no evidence for universal prediction of whole interactions or individual behavior. Strong prediction is achievable for constrained motor and choice processes, routine skilled performance, and near-term transitions when the task, state, opportunity, and method are specified. Discovery, errors, recovery, and abandonment are predictively tractable as probabilistic event processes. Adaptation, expertise, transfer, and strategy can be forecast only through richer longitudinal and latent-state models, and their individual trajectories remain substantially uncertain.

The warranted contribution is a predictive architecture rather than a new predictive primitive. It separates prediction from explanation and intervention; defines target, horizon, information, uncertainty, and validity; and requires held-out evaluation, calibration, transport tests, and explicit failure conditions. Interaction becomes predictively scientific when forecasts are reproducible, uncertainty-bearing, and bounded by declared contracts. Where those conditions cannot be met, the correct result is not a confident point estimate but a broad distribution, an incomparable context, or a statement that the target is not identified.

Reusable Artifacts

A. Prediction Contract

Field	Required declaration
Identity	Prediction identifier, version, date, owner, intended scientific or operational use.
Target	Outcome, unit, level, horizon, event definition, censoring, competing events.
Information cutoff	Timestamp or episode boundary beyond which no input may be used.
Inputs	Observed records, derived variables, latent estimates, provenance, measurement uncertainty.
Interaction scope	Participants, episode boundaries, system state, projection, opportunity, policy, context.
Model	Family, parameters, priors, regularization, training procedure, update rule.
Causal status	Associational, interventional, transported, mechanism-based, or policy-value claim.
Prediction form	Probability, distribution, interval, set, hazard, sequence, ranking, or scenario.
Uncertainty	Aleatoric, epistemic, measurement, state, structural, distributional, reflexive.
Evaluation	Baselines, holdout design, metrics, calibration, coverage, utility, subgroup tests.
Validity domain	Population, task, projection, modality, system, policy, environment, time.
Transport conditions	Assumptions and evidence required for use outside the evaluation domain.
Failure conditions	Minimum gain, maximum error, calibration tolerance, abstention rule, stop-use conditions.
Monitoring	Drift tests, recalibration schedule, update triggers, audit trail.

B. Prediction Evidence Hierarchy

Level	Evidence type	Meaning
0	Raw record	Timestamp, event payload, sensor sample.
1	Verified observation	Quality-controlled event or state measurement.
2	Derived predictor	Latency, path length, opportunity-set difference.
3	Latent estimate	Belief, strategy posterior, expertise score.
4	In-sample association	Fit within data used to construct the model.
5	Internal held-out prediction	Performance on held-out units from the same regime.
6	Temporal or group generalization	Prediction for future periods or unseen participants.
7	Cross-context transport	Prediction across tasks, projections, devices, or systems.
8	Interventional prediction	Forecast under a deliberately changed condition.
9	Replicated predictive regularity	Independent replications with stable scope and uncertainty.

C. Predictive Metric Registry

Target form	Metrics	Interpretation	Example
Continuous outcome	MAE/RMSE	Magnitude error; RMSE emphasizes large errors.	Completion time, search time.
Probabilistic binary	Log loss/Brier score	Proper scoring of probability forecasts.	Error or abandonment probability.
Ranking	ROC-AUC/PR-AUC	Discrimination; PR-AUC preferred for rare positives.	Risk ranking.
Calibration	Calibration slope/intercept, ECE, reliability curve	Agreement of probabilities and frequencies.	All probabilistic targets.
Intervals/sets	Coverage and width/set size	Validity and sharpness.	Time interval or action-set prediction.
Survival	C-index, integrated Brier score, time calibration	Censoring-aware event prediction.	Discovery, recovery, abandonment.
Sequence	Cross-entropy, edit distance, horizon accuracy	Next-event and trajectory prediction.	Action sequence.
Decision	Expected utility, net benefit, regret	Value under an intervention threshold or policy.	Adaptive help.
Transport	Performance delta and recalibration	Stability across declared domains.	GUI to voice; novice to expert.

	need		
Abstention	Risk-coverage curve	Quality when model may decline to predict.	High-uncertainty episodes.

D. Uncertainty Model

Type	Definition	Required treatment
Aleatoric	Outcome variability remaining under the information set.	Predictive distribution; irreducible risk.
Epistemic	Limited evidence about model or parameters.	Posterior, ensemble, bootstrap; decreases with relevant data.
Measurement	Instrument, coding, segmentation, construct error.	Propagate from measurement contract.
State	Hidden or partially observed current conditions.	Belief state or filtering distribution.
Structural	Competing models or causal graphs.	Model averaging or sensitivity analysis.
Distributional	Shift in population, task, projection, policy, or time.	Domain tests, drift monitoring, robust sets.
Reflexive	Prediction or intervention changes behavior.	Policy-aware evaluation and experimentation.

E. Prediction Protocol Checklist

1. Declare target: Define the outcome, unit, horizon, event time, censoring, and competing outcomes.
2. Freeze information cutoff: List every variable available at prediction time; prohibit future-derived features.
3. State use: Separate scientific forecasting, diagnosis, causal intervention, control, and operational triage.
4. Specify population and context: Declare users or agents, task, projection, device, authority, environment, and time period.
5. Register baselines: Include prevalence, persistence, simple rules, human judgment, and established task models.
6. Select evaluation design: Use temporal, participant, task, projection, or deployment holdout appropriate to the intended claim.
7. Fit without leakage: Keep preprocessing, feature selection, imputation, and tuning within training folds.
8. Quantify uncertainty: Produce calibrated probabilities, intervals, sets, or hazards with named guarantees.
9. Evaluate multiple properties: Report discrimination, calibration, coverage, sharpness, utility, subgroup and context performance.
10. Test transport: Evaluate across time, participants, tasks, projections, systems, and policies when those claims are intended.
11. Test interventions separately: Use experiments or identified causal models for claims about changing outcomes.
12. Monitor and update: Detect drift, recalibrate, revise the validity domain, and document model changes.
13. State failure conditions: Define minimum acceptable gain, calibration, coverage, and decision utility before use.
14. Preserve evidence lineage: Link each prediction to the interaction, behavior, and measurement contracts that generated its inputs.

F. Open Problems

- Predictive identification of strategies under observational equivalence.
- Calibrated individual prediction under sparse, nonstationary histories.
- Transport across projections that preserve semantics but alter opportunity and measurement.
- Prediction under adaptive interfaces where the model changes the data-generating process.
- Causal prediction of assistance, explanation, and recovery interventions.
- Joint uncertainty propagation from raw records through latent constructs to outcomes.
- Benchmarks that preserve interaction boundaries and prevent target leakage.

- Evaluation of whether the program's typed constructs add out-of-domain predictive gain.

References

- Anderson, J. R. (2007). *How Can the Human Mind Occur in the Physical Universe?* Oxford University Press.
- Angelopoulos, A. N., & Bates, S. (2021). *A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification*. arXiv:2107.07511.
- Card, S. K., Moran, T. P., & Newell, A. (1983). *The Psychology of Human-Computer Interaction*. Erlbaum.
- Card, S. K. (1986). The Model Human Processor: A Model for Making Engineering Calculations of Human Performance. *Proceedings of the Human Factors Society*.
- Fitts, P. M. (1954). The Information Capacity of the Human Motor System in Controlling the Amplitude of Movement. *Journal of Experimental Psychology*, 47(6), 381-391.
- Hick, W. E. (1952). On the Rate of Gain of Information. *Quarterly Journal of Experimental Psychology*, 4(1), 11-26.
- Hyman, R. (1953). Stimulus Information as a Determinant of Reaction Time. *Journal of Experimental Psychology*, 45(3), 188-196.
- Jastrzemski, T. S., & Charness, N. (2007). The Model Human Processor and the Older Adult: Parameter Estimation and Validation Within a Mobile Phone Task. *Journal of Experimental Psychology: Applied*, 13(4), 224-248.
- Kaelbling, L. P., Littman, M. L., & Cassandra, A. R. (1998). Planning and Acting in Partially Observable Stochastic Domains. *Artificial Intelligence*, 101, 99-134.
- Lei, J., G'Sell, M., Rinaldo, A., Tibshirani, R. J., & Wasserman, L. (2018). Distribution-Free Predictive Inference for Regression. *Journal of the American Statistical Association*, 113(523), 1094-1111.
- Pearl, J. (2009). *Causality: Models, Reasoning, and Inference* (2nd ed.). Cambridge University Press.
- Pearl, J. (2012). The Do-Calculus Revisited. *Proceedings of UAI*.
- Shmueli, G. (2010). To Explain or to Predict? *Statistical Science*, 25(3), 289-310.
- Sutton, R. S., & Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press.
- Yarkoni, T., & Westfall, J. (2017). Choosing Prediction Over Explanation in Psychology: Lessons From Machine Learning. *Perspectives on Psychological Science*, 12(6), 1100-1122.
- Earlier papers in the Computational Interaction research program: Interaction; Behavior; Measurement; Representation, Invariance, and Projection; Opportunity, Affordance, and Action Availability; Interaction Similarity; Interaction Transfer; Interaction Strategies; Formal Models of Computational Interaction.